

Day – 14 Build and configure a small office network deliverable.

Task – 1 Assign ip to 5 devices.

- Step 1: Design the Subnet Scheme
- Before touching any device, define your network parameters to ensure all devices reside in the exact same broadcast domain and subnet:
- Network Subnet: 192.168.10.0
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.10.1 (Router interface)
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- Step 2: Assign Static IP Addresses to the 5 Devices
- Configure each device with a unique IP address within the same subnet, keeping them outside of any dynamic DHCP pool range to avoid conflicts:
- Workstation 1 (PC0 - Admin Laptop):
- IP Address: 192.168.10.10
- Subnet Mask: 255.255.255.0
- Gateway: 192.168.10.1
- Workstation 2 (PC1 - Lab Laptop):
- IP Address: 192.168.10.11
- Subnet Mask: 255.255.255.0
- Gateway: 192.168.10.1
- Workstation 3 (PC2 - Desktop A):
- IP Address: 192.168.10.12
- Subnet Mask: 255.255.255.0
- Gateway: 192.168.10.1
- Workstation 4 (PC3 - Desktop B):
- IP Address: 192.168.10.13
- Subnet Mask: 255.255.255.0
- Gateway: 192.168.10.1
- Workstation 5 (PC4 - Test Server):
- IP Address: 192.168.10.14
- Subnet Mask: 255.255.255.0
- Gateway: 192.168.10.1
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- Step 3: Verify Connectivity Using the ping Command

- Once all five devices are connected to the same switch and configured with their static IPs, test layer-3 reachability from your main administrative laptop (192.168.10.10):
- Open the command prompt on PC0.
- Ping each device sequentially to ensure bidirectional communication:
- DOS
- ping 192.168.10.11
- ping 192.168.10.12
- ping 192.168.10.13
- ping 192.168.10.14
- Expected Output: A successful reply (Reply from 192.168.10.x: bytes=32 time<1ms TTL=128) with 0% packet loss confirms that all devices are properly communicating within the same static subnet.

Task – 2 Setup DHCP server with automatically ip assign.

- Here is how you can set up a DHCP server to automatically lease unique addresses to at least three devices:
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- Step 1: Configure the Server's Static IP
- Before a server can distribute IP addresses to others, it must have a fixed, permanent address itself.
- Place a Server (Server0) and connect it to a Switch, alongside three client devices (PC0, PC1, and Laptop0).
- Click on Server0, go to the Desktop tab > IP Configuration, and set a static IP:
- IP Address: 192.168.20.2
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.20.1
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- Step 2: Set Up the DHCP Pool and Scope
- Still on Server0, navigate to the Services tab and select DHCP.
- Turn the DHCP service ON.
- Create or modify the default pool (e.g., name it Office-Pool) with the following parameters:
- Default Gateway: 192.168.20.1
- DNS Server: 8.8.8.8 (or your local DNS server)
- Start IP Address: 192.168.20.100
- Maximum Number of Users: 50 (This creates a pool range from 192.168.20.100 to 192.168.20.149)
- Click Add (or Save) to commit the pool settings.

- Step 3: Connect and Configure the Three Client Devices
- Now, configure your client devices to request their configuration dynamically:
- Click on PC0 > Desktop > IP Configuration, and switch from Static to DHCP. Watch as it completes the DORA handshake and populates its fields.
- Repeat the same process for PC1.
- Repeat the same process for Laptop0.
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- Step 4: Verify Assigned Unique IPs from the Correct Range
- To confirm that each device successfully received a unique IP address from your specified range, open the command prompt on each device and run ipconfig:
- PC0 Assigned IP: 192.168.20.100
- PC1 Assigned IP: 192.168.20.101
- Laptop0 Assigned IP: 192.168.20.102

Task – 3 Config the DNS server.

- Here is how you can set this up using a network-wide DNS tool like dnsmasq (ideal for making it work across *all* connected devices) or via a local hosts file for a quick test.
- If you have a local Linux server, a Raspberry Pi, or a lab machine running on your network, dnsmasq is a lightweight, easy-to-configure DNS forwarder that lets you assign custom local hostnames instantly.
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- Step 1: Install dnsmasq
- On your local DNS server machine, install dnsmasq:
- Bash
- `sudo apt update`
- `sudo apt install dnsmasq`
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- Step 2: Configure the Custom Hostname Mapping
- Open the dnsmasq configuration directory or main config file:
- Bash
- `sudo nano /etc/dnsmasq.d/custom-domains.conf`
- Add an address mapping line that points your custom hostname to your target device's local IP address (for example, 192.168.1.100):
- `address=/musicplayer.local/192.168.1.100`
- Save and close the file.
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- Step 3: Restart the Service
- Apply your new configuration by restarting the dnsmasq service:

- Bash
- `sudo systemctl restart dnsmasq`
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- Step 4: Point Other Devices to Your DNS Server
- For other devices on your network to resolve `musicplayer.local`, update your router's DHCP settings (or the individual device's network settings) so that the Primary DNS Server points to the IP address of the machine running `dnsmasq`.

Task – 4 Create two VLANs.

- Here is how you can set up two VLANs in a network simulator like Cisco Packet Tracer and verify their isolation:
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- Step 1: Create the VLANs on the Managed Switch
- Access your switch's command-line interface (CLI) to create VLAN 10 and VLAN 20:
- `Switch> enable`
- `Switch# configure terminal`
- `Switch(config)# vlan 10`
- `Switch(config-vlan)# name Workstations`
- `Switch(config-vlan)# exit`
- `Switch(config)# vlan 20`
- `Switch(config-vlan)# name Smart_Devices`
- `Switch(config-vlan)# exit`
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- Step 2: Assign Switch Ports to the VLANs
- Assign specific physical access ports on the switch to your new VLANs based on where your devices are plugged in:
- Assign Ports for VLAN 10 (Workstations):
- `Switch(config)# interface range fastEthernet 0/1 - 2`
- `Switch(config-if-range)# switchport mode access`
- `Switch(config-if-range)# switchport access vlan 10`
- `Switch(config-if-range)# exit`
- Assign Ports for VLAN 20 (Smart Devices):
- `Switch(config)# interface range fastEthernet 0/3 - 4`
- `Switch(config-if-range)# switchport mode access`
- `Switch(config-if-range)# switchport access vlan 20`
- `Switch(config-if-range)# exit`
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- Step 3: Configure Device IP Addresses

- Connect your devices to the designated switch ports and assign static IP addresses corresponding to their respective subnets:
- VLAN 10 ('Workstations' Subnet - 192.168.10.x):
- PC0 (Port Fa0/1): IP 192.168.10.10, Subnet Mask 255.255.255.0
- PC1 (Port Fa0/2): IP 192.168.10.11, Subnet Mask 255.255.255.0
- VLAN 20 ('Smart Devices' Subnet - 192.168.20.x):
- Smart-Device0 (Port Fa0/3): IP 192.168.20.20, Subnet Mask 255.255.255.0
- Smart-Device1 (Port Fa0/4): IP 192.168.20.21, Subnet Mask 255.255.255.0
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- Step 4: Test Connectivity (Intra-VLAN vs. Inter-VLAN)
- Open the command prompt on PC0 (VLAN 10) and run your verification pings:
- Test Same VLAN Communication:
- DOS
- ping 192.168.10.11
- Result: Successful replies. Devices in the same VLAN can freely communicate because they belong to the same broadcast domain and subnet.
- Test Cross-VLAN Isolation:
- DOS
- ping 192.168.20.20
- Result: Request timed out (100% packet loss). Even though the devices are plugged into the exact same physical switch, Layer 2 isolation prevents traffic from crossing between VLAN 10 and VLAN 20 without an active router or Layer 3 switch.

Task – 5 Based on scenario.

- Configuring Network Address Translation (NAT) on a router is what bridges the gap between your private local area network (LAN) and the public internet, allowing multiple devices sharing private IP addresses to communicate using a single public IP.
- Here is how you can set up and verify NAT in a network simulator like Cisco Packet Tracer:
- Step 1: Define Your Interfaces (Inside and Outside)
- First, log into your edge router (e.g., Router0) via its CLI to designate which interface connects to your private LAN and which faces the public internet/ISP:
- Router> enable
- Router# configure terminal
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- ; Configure the LAN-facing interface (Inside)
- Router(config)# interface gigabitEthernet 0/0
- Router(config-if)# ip address 192.168.1.1 255.255.255.0
- Router(config-if)# ip nat inside

- Router(config-if)# no shutdown
- Router(config-if)# exit
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- ; Configure the WAN-facing interface (Outside)
- Router(config)# interface gigabitEthernet 0/1
- Router(config-if)# ip address 203.0.113.2 255.255.255.0
- Router(config-if)# ip nat outside
- Router(config-if)# no shutdown
- Router(config-if)# exit
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- Step 2: Configure NAT Overload (PAT) and Access Control
- To allow multiple private IP addresses to translate through a single public IP address, configure Port Address Translation (PAT) using an access control list:
- ; Define which local subnet is permitted to be translated
- Router(config)# access-list 1 permit 192.168.1.0 0.0.0.255
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- ; Bind the access list to the outside interface overload pool
- Router(config)# ip nat inside source list 1 interface gigabitEthernet 0/1 overload
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- Step 3: Set Up a Default Route to the ISP
- Your router needs to know where to send packets destined for the outside world when it doesn't have a specific local path:
- Router(config)# ip route 0.0.0.0 0.0.0.0 203.0.113.1
- *(Where 203.0.113.1 is the simulated ISP next-hop gateway IP address).*
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- Step 4: Test Internet Access from a LAN Client
- Connect a client PC (PC0) to the LAN switch, ensuring its IP configuration points to 192.168.1.1 as its Default Gateway and has a valid DNS server configured.
- Open the command prompt on PC0 and test connectivity to an external simulated server or public address (such as www.wikipedia.org or a public IP like 8.8.8.8):
- DOS
- ping 203.0.113.50
- Verify NAT Translation: On the router CLI, run the following command to watch translations happen in real-time as your client communicates outward:
- Router# show ip nat translations
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